```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans, DBSCAN

from prophet import Prophet
from sklearn.cluster import DBSCAN
```

[1] ✓ 3.4s Python

```
c:\Users\rakhi\OneDrive\Desktop\infosys-spring board\.venv\Lib\site-packages\tqdm\auto.py:21: TqdmWarning: IProgress not found. Please update jupyter and ipywidgets. See https://ipywidgets.readthedocs.io/en/stable/user_insta
  from .autonotebook import tqdm as notebook_tqdm
```

```python
df = pd.read_csv("../outputs/processed_data.csv")

df['timestamp'] = pd.to_datetime(df['timestamp'])
df.sort_values('timestamp', inplace=True)
```

[7] ✓ 0.0s Python

```python
window = 10

df['hr_mean'] = df['heart_rate_bpm'].rolling(window).mean()
df['hr_std'] = df['heart_rate_bpm'].rolling(window).std()

df['steps_mean'] = df['steps'].rolling(window).mean()
df['steps_std'] = df['steps'].rolling(window).std()

df['spo2_mean'] = df['spo2_pct'].rolling(window).mean()
df['spo2_std'] = df['spo2_pct'].rolling(window).std()

df.fillna(0, inplace=True)
```

[8] ✓ 0.0s Python

```python
features = df[['heart_rate_bpm','steps','spo2_pct',
               'hr_mean','hr_std',
               'steps_mean','steps_std',
               'spo2_mean','spo2_std']]
```

✓ 0.0s Python

```python
scaler = StandardScaler()
X = scaler.fit_transform(features)
```

✓ 0.0s Python

```python
dbscan = DBSCAN(eps=0.5, min_samples=5)

df['dbscan'] = dbscan.fit_predict(X)

print("DBSCAN outliers:", (df['dbscan'] == -1).sum())
```

✓ 5.6s Python

```
DBSCAN outliers: 373
```

```python
features.to_csv("../outputs/feature_matrix.csv", index=False)
```

✓ 0.0s Python

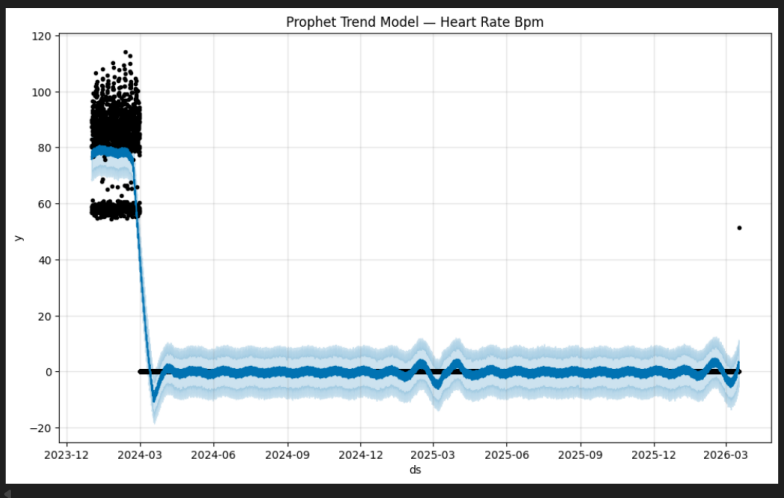
```python
prophet_df = df[['timestamp','heart_rate_bpm']].rename(
    columns={'timestamp':'ds','heart_rate_bpm':'y'}
)

model = Prophet()
model.fit(prophet_df)

future = model.make_future_dataframe(periods=0)
forecast = model.predict(future)

fig = model.plot(forecast)
plt.title("Prophet Trend Model — Heart Rate Bpm")

plt.savefig("../outputs/prophet_heart_rate_bpm.png")
```

✓ 30.9s Python

```python
plt.savefig("../outputs/prophet_heart_rate_bpm.png")
plt.show()
```

✓ 30.9s

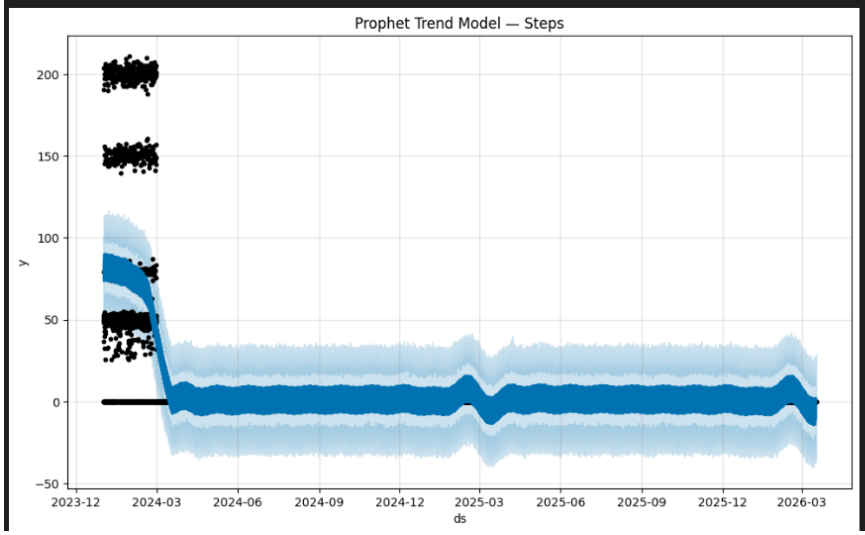
```
16:28:09 - cmdstanpy - INFO - Chain [1] start processing
16:28:35 - cmdstanpy - INFO - Chain [1] done processing
```



```python
prophet_df = df[['timestamp','steps']].rename(
    columns={'timestamp':'ds','steps':'y'}
)

model = Prophet()
model.fit(prophet_df)

future = model.make_future_dataframe(periods=0)
forecast = model.predict(future)

fig = model.plot(forecast)
plt.title("Prophet Trend Model — Steps")

plt.savefig("../outputs/prophet_steps.png")
plt.show()
```

✓ 18.2s

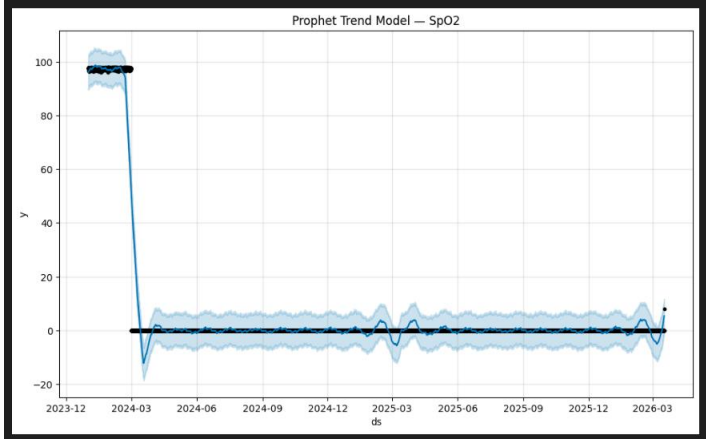
```
16:28:43 - cmdstanpy - INFO - Chain [1] start processing
16:28:56 - cmdstanpy - INFO - Chain [1] done processing
```




```python
    columns={'timestamp':'ds','spo2_pct':'y'}
)

model = Prophet()
model.fit(prophet_df)

future = model.make_future_dataframe(periods=0)
forecast = model.predict(future)

fig = model.plot(forecast)
plt.title("Prophet Trend Model — SpO2")

plt.savefig("../outputs/prophet_spo2_pct.png")
plt.show()
```

✓ 23.1s

```
16:29:04 - cmdstanpy - INFO - Chain [1] start processing
16:29:23 - cmdstanpy - INFO - Chain [1] done processing
```



```python
scaler = StandardScaler()
X = scaler.fit_transform(features)
```

[36] ✓ 0.0s Python

```python
kmeans = KMeans(n_clusters=4, random_state=42)
df['cluster'] = kmeans.fit_predict(X)

print("KMeans clusters:", np.unique(df['cluster']))
```

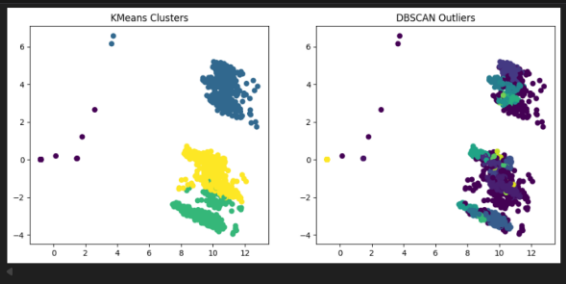
[37] ✓ 3.0s Python

```
KMeans clusters: [0 1 2 3]
```

```python
dbscan = DBSCAN(eps=0.5, min_samples=5)
df['dbscan'] = dbscan.fit_predict(X)

print("DBSCAN outliers:", (df['dbscan'] == -1).sum())
```

[38] ✓ 4.0s Python

```
DBSCAN outliers: 373
```

```python
from sklearn.decomposition import PCA

pca = PCA(n_components=2)
X_pca = pca.fit_transform(X)

plt.figure(figsize=(12,5))

# KMeans plot
plt.subplot(1,2,1)
plt.scatter(X_pca[:,0], X_pca[:,1], c=df['cluster'])
plt.title("KMeans Clusters")

# DBSCAN plot
plt.subplot(1,2,2)
plt.scatter(X_pca[:,0], X_pca[:,1], c=df['dbscan'])
plt.title("DBSCAN Outliers")

plt.savefig("../outputs/milestone2_clusters.png")
plt.show()
```

[39] ✓ 2.6s Python